

Day 15 lab: capstone build and demonstrations

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- ▶ **Goal:** a working capstone system and a short report.
- ▶ **Boards:** the XIAOMI Kits and the Raspberry Pi 5 boards of your team.
- ▶ **Deliverable:** the demonstration of 10 minutes and the report `report.md`.

The last day

After Part 1 of the lecture, the lecture time (Parts 2 and 3, 100 minutes) is build time. The instructor visits each team. Then the lab has 80 minutes of build and measurement, the demonstrations, and the close. `Docs/capstone.md` gives the requirements and the rubric.

Key point: Finish the measurements before the polish. A working system with numbers scores more than a nice system with no numbers.

Plan of the three hours



Part	Work	Time
A	Build: complete the system and the measurements	80 min
B	Demonstrations: 10 minutes for each team	70 min
C	Close: submit the report, give course feedback	30 min

The order of the demonstrations is the schedule of Day 14.

Hardware for each team

- ▶ Two XIAOMI Kits and two Raspberry Pi 5 boards with the cameras, the cards, and the 27 W supplies.
- ▶ Two or more laptops in the lab network.
- ▶ Optional: a USB power meter for the energy (Day 9).
- ▶ The instructor keeps spare boards ready.

File	Content
report.md	The report template. Fill it, hand it in.
feedback.md	The course feedback, with no name
schedule.py	The instructor: the order of the demonstrations
solutions/ TEST_NOTES.md	An example report Test steps for the instructor

Lab files	Labs/day15/ and the lab folders of Days 1 to 14
Your plan	The plan for Day 15 in your Day 14 report
The brief	Docs/capstone.md: requirements, rubric, report

Part A: build and measure (80 min)

1. **Complete the system (40 min).** Connect the parts. Test each requirement of your proposal once.
2. **Measure (30 min).** Fill the benchmark table of `report.md`. Measure the size and the accuracy of the model before and after the optimization on the same test set.
3. **Rehearse (10 min).** Run the demonstration once with the clock: the live system, the cut of the network link, the dashboard. Prepare plan B: a recorded input or a short video.

Before Part A

The lecture time (100 minutes) is build time. Build the parts that the demonstration needs first. Follow the plan of your Day 14 report.

Model	Board	Format	Median	p95	Memory	Energy
...	XIAO	int8	... ms	... ms	arena ... bytes	... mJ
...	Pi 5	...	... ms	... ms	RSS ... MB	... mJ

The method of Day 9, in the report under the table:

- ▶ warm-up runs and the number of timed runs,
- ▶ the window: the model only, or end to end,
- ▶ the statistic: the median and a high percentile,
- ▶ the energy: a USB power meter, or the power values of Day 9 (say “estimate”).

Source: The form of the table in `Labs/day15/report.md`. The values come from your boards.

Part A: the five requirements, a last check

No.	Requirement	Show it in the demonstration
1	XIAOML Kit and Raspberry Pi	Each board with its own task
2	One optimized model, size and accuracy before and after	The table of Section 4 of the report
3	One local decision, no internet	Cut the link to the instructor broker; the decision continues
4	Telemetry with MQTT to a dashboard	The live dashboard
5	Benchmark table: latency, memory, energy	The table of the previous frame

A requirement that does not work: show what works, use plan B, and write the limit in the report.

Part B: the demonstration (10 min)

Time	Content
1 min	The problem and the requirements
4 min	The system works: live, also with the network link cut
3 min	The benchmark table, two decisions with their numbers
2 min	Questions

- ▶ Start on time: the next team waits.
- ▶ Show, then explain. The live system comes before the slides.
- ▶ Answer with numbers.
- ▶ Watch the other teams. Write one question for each team.

Part B has 70 minutes: 7 teams with 10 minutes each. With more teams, the instructor uses the schedule of Day 14.

Criterion	Weight	Excellent
The system works in the demonstration	30 %	All five requirements work live, also with the link cut
Integration of the two boards, messages, and monitoring	25 %	Each board has a clear task; topics and sequence numbers; a live dashboard
Measured results and analysis of trade-offs	25 %	A complete table with the method; each main decision with two measured numbers and what you gave up
Report and presentation	20 %	Short, clear, the template; answers with numbers

Docs/capstone.md, Section 3, gives the levels “adequate” and “insufficient”.

Source: Weights: Section 7 of the syllabus. Levels of the third criterion: Decision Log rubric of the textbook instructor guide.

The report: at most four pages

1. Problem and requirements
2. The system
3. The five requirements
4. The optimized model
5. The benchmark table
6. Robustness
7. Decision Log (two or three)
8. Limits and next steps

The example report of the machine monitor is in `solutions/report_example.md`.

- ▶ **Report (20 min):** complete `report.md`, and submit it as the instructor says.
- ▶ **Feedback (10 min):** fill `feedback.md` with no name: eight ratings, the level of each week, four open questions.

The instructor grades with the capstone rubric (Docs/capstone.md, Section 3):

- ▶ ✓ The system works in the demonstration (30 percent).
- ▶ ✓ Integration of the two boards, messages, and monitoring (25 percent).
- ▶ ✓ Measured results and analysis of trade-offs (25 percent).
- ▶ ✓ Report and presentation (20 percent).

Hand in: `report.md`, and `feedback.md` with no name.

Which decision of your system had the largest effect on the result, and which measured number shows it?

The report has two or three Decision Logs of about 100 words each. A good one:

- ▶ states the decision and the options,
- ▶ gives at least two measured numbers with their method,
- ▶ names the trade-off: what you gained and what you gave up,
- ▶ says which numbers are estimates.

Problem	Fix
A part does not work in Part A	Show the parts that work and plan B; write the limit in the report
The live demonstration fails	Use plan B: a recorded input or a video. Explain the cause.
The benchmark table has gaps	Measure the main model first; write “not measured” in a gap
The energy has no meter	Use the power values of Day 9 and say “estimate”
The demonstration takes more than 10 minutes	Rehearse with the clock in Part A, step 3
The README of the lab has the steps and the schedule commands.	

This lecture adapts material from these sources:

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